Curriculum Vitae

(July 24, 2026)

Sangwon Kim

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Research Interests

· Ultrafast tunneling in nanocontacts
· Quasiclassical quantum theory
· Bohmian mechanics
· Quantum optics

Education

2019/02/25 - 2026/08/21	(Scheduled) Ph.D. in Physics, Korea Advanced Institute of Science and Technology, Korea (Thesis: Quasiclassical theory of nonadiabatic tunneling in nanocontacts driven by classical and quantum light)
2014/03/01 - 2019/02/24	B.S. in Physics, Korea Advanced Institute of Science and Technology, Korea

Awards/Appreciation

May 2024	Best Student Paper Award ALTA 2024

Publication List

3. (Submitted) Electronic Switching at the Atomic Spatio-Temporal Scale
Marwin Gedamke, Sebastian Grossenbach, Sven Väth, Matthias Falk, Stefan Eggert, **Sangwon Kim**, Andrey S. Moskalenko, Daniele Brida, Markus Ludwig, and Alfred Leitenstorfer

2. (Submitted) Tunneling driven by quantum light described via field Bohmian trajectories
Sangwon Kim, Seongjin Ahn, Denis V. Seletskiy, and Andrey S. Moskalenko
arXiv:quant-ph/2507.15972

1. Quasiclassical theory of non-adiabatic tunneling in nanocontacts induced by phase-controlled ultrashort light pulses
Sangwon Kim, Tobias Schmude, Guido Burkard, and Andrey S. Moskalenko
New. J. Phys. 23 083006 (2021). [arXiv:cond-mat.mes-hall/2011.15125]

Presentations

2. QAMTS 2024 at Donostia-San Sebastián, Spain (June 2024)
   Quasiclassical approach to dynamical aspects of tunneling in nanocontacts induced by ultrashort light pulses
1. OPC 2021 at Jeju, Korea (Jul. 2021)
   Quasiclassical theory of non-adiabatic tunneling in nanocontacts induced by ultrashort light pulses

Conferences/Meetings Participated

9. (Scheduled) UP 2026
   Sapporo, Japan (August 2 - 7, 2026)
8. ALTA 2026
   Jeju, Korea (May 6 - 9, 2026)
7. QAMTS 2024
   Donostia-San Sebastián (Basque Country), Spain (June 16 - 21, 2024)
6. ALTA 2024
   Jeju, Korea (May 1 - 4, 2024)
5. GRC Ultrafast Phenomena in Cooperative Systems
   Lucca (Barga), Italy (February 4 - 9, 2024)
4. ALTA 2023
   Jeju, Korea (May 1 - 4, 2003)
3. ICAMD 2021
   Jeju, Korea (December 6 - 10, 2021)
2. OPC 2021
   Jeju, Korea (July 4 - 7, 2021)
1. GRS/GRC Ultrafast Phenomena in Cooperative Systems
   Lucca (Barga), Italy (February 1 - 7, 2020)

Teachings

14. Spring 2026, Quantum Optics (PH624), TA, KAIST
13. Fall 2025, General Physics I (PH141), Vice Head TA, KAIST
12. Spring 2025, General Physics II (PH142), TA, KAIST
11. Fall 2024, General Physics II (PH142), TA, KAIST
10. Fall 2023, Graduate Quantum Mechanics II (PH504), TA, KAIST
9. Spring 2023, Graduate Classical Mechanics (PH505), TA, KAIST
8. Fall 2022, Physics Experiment I (PH251), TA, KAIST
7. Spring 2022, Quantum Optics (PH624), TA, KAIST
6. Fall 2021, Nonlinear Optics (PH721), TA, KAIST
5. Spring 2021, Graduate Classical Mechanics (PH505), TA, KAIST
4. Fall 2020, Classical Mechanics II (PH222), TA, KAIST
3. Spring 2020, Classical Mechanics I (PH221), TA, KAIST
2. Fall 2019, Experiment Oriented General Physics II (PH172), TA, KAIST
1. Spring 2019, General Physics II (PH142), TA, KAIST

Other Activities

2. Visited femtoQ, Prof. Denis V. Seletskiy Group, Polytechnique Montréal
   Montréal, Canada (December 5 - December 20, 2025).

1. Visited Prof. Guido Burkard Group and Prof. Alfred Leitenstorfer Group, Universität Konstanz
   Konstanz, Germany(January 7 - February 17, 2020).

Skills

1. Professional working proficiency in English.
2. Proficient in Mathematica, Maple, and Python/Jupyter.
3. Experienced with Linux Bash and C.
4. Completed all coursework required for a Bachelor's degree in Electrical Engineering, although the degree was not formally awarded.
5. Fully audited (participated all course activies including homeworks and exams) the following courses:
   · Modern Algebra I (MAS311)